being undercharged. The reason for undercharging may be an improperly adjusted voltage regulator, but most likely it is the result of voltage drop in the circuit caused either by inadequate wiring and connections (see above) or else *by installing isolation diodes without compensating for the voltage drop of 0.5 to 1.0 volt caused by the diodes* (see Chapter 1 for an explanation of this; for solutions to the problem, see pages 107–9).

- Check all the connections in any external field circuit (see page 106) to make sure they are clean and tight.

Only suspect the alternator if none of the preceding tests turns up a problem. One or more diodes or stator windings may be open-circuited (pages 99, 104), or the brushes may be worn or not making good contact with their slip rings, in which case output is likely to be erratic (page 100). Such problems are uncommon—an alternator is far more likely to fail outright than it is to partially fail. (Remember, this section is about persistent undercharging *from an otherwise functioning alternator*; for no output, see below).

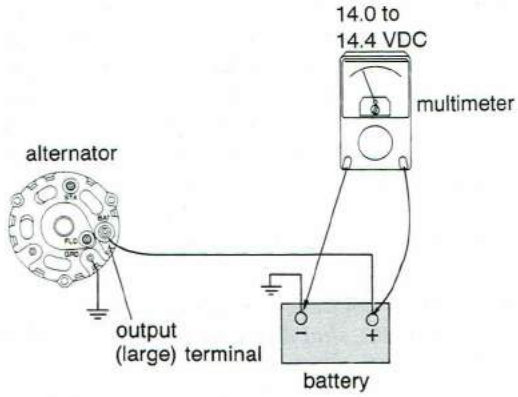


FIGURE 3-30A. Testing for the maximum voltage to which an alternator is driving a battery during charging.



FIGURE 3-30B. Checking the absorption voltage at the battery.

Troubleshooting Chart 3-1.
Battery/Alternator/Regulator Problems: Undercharging

Symptom: Battery never seems to regain full charge.

Is the battery dying or dead? **NO** **TEST:** Check the specific gravity or open-circuit voltage. Charge up and run a load test or capacity test.	**YES** **FIX:** Look for equipment inadvertently left on, dirt or moisture on the battery, or a leak to ground in the boat's wiring. Run an equalization charge on wet-cells; try the dead-battery recovery procedure on gel-cells and AGMs. If battery fails to respond, replace it.
Is belt tension adequate? **NO** **TEST:** Listen for belt squeal. Watch for fluctuating tachometer. Depress the belt—it should not move more than 3/8 to 1/2 inch under moderate finger pressure.	**YES** **FIX:** Increase tension.
Is the alternator undersized or its speed of rotation too slow? Is charging time inadequate? **NO** **TEST:** Compare demands on alternator with its size and speed of rotation as outlined in Chapter 1. Try charging the battery for at least 3 hours.	**YES** **FIX:** Match alternator with demands (see Chapter 1). Increase routine charging time.
Is there excessive voltage drop in the circuit between the alternator and the battery? **NO** **TEST:** Use a multimeter to test for volt drop *at full alternator output* (during the bulk charge phase) with one meter probe on the alternator positive stud and the other on the battery positive post, and then one probe on the alternator negative and the other on the battery negative post.	**YES** **FIX:** Clean and tighten loose, dirty, and corroded terminals. (Be sure to check the engine ground-strap connections.) Replace faulty or undersized wiring. Remove battery isolation diodes and replace with a battery paralleling relay or adjust the voltage regulator for diode-induced voltage drop.
Is the voltage at the battery too low for effective charging? **NO** **TEST:** As the battery comes to charge (reaches the absorption voltage), monitor the voltage across the battery terminals. Does it stabilize below 14.0 volts?	**YES** **FIX:** Adjust the voltage regulator upward. Compensate for diode-induced voltage drop (see the text). If this is not possible, get rid of battery isolation diodes and/or replace the voltage regulator with a more appropriate model.
Is the voltage regulator wiring defective? **NO** **TEST:** Check external connections and wiring.	**YES** **FIX:** Repair or replace.
If the above tests fail to reveal a problem, check the brushes, diodes, rotor, and stator windings in the alternator (see pages 99–104).	